

```
In [1]: import pandas as pd
```

```
In [2]: data = {
    'age': [25, 30, 22, 40, 55, 60, 33, 28, 45, 50],
    'income': [50000, 60000, 45000, 70000, 80000, 90000, 65000, 62000, 75000, 85000],
    'age_group': ['20-30', '30-40', '20-30', '40-50', '50-60', '50-60', '30-40', '20-30', '40-50', '50-60']
}
```

```
In [3]: df = pd.DataFrame(data)
```

```
In [4]: df
```

Out[4]:

	age	income	age_group
0	25	50000	20-30
1	30	60000	30-40
2	22	45000	20-30
3	40	70000	40-50
4	55	80000	50-60
5	60	90000	50-60
6	33	65000	30-40
7	28	62000	20-30
8	45	75000	40-50
9	50	85000	50-60

```
In [7]: df.groupby('age_group')['income'].describe()
```

Out[7]:

	count	mean	std	min	25%	50%	75%	max
age_group								
20-30	3.0	52333.333333	8736.894948	45000.0	47500.0	50000.0	56000.0	62000.0
30-40	2.0	62500.000000	3535.533906	60000.0	61250.0	62500.0	63750.0	65000.0
40-50	2.0	72500.000000	3535.533906	70000.0	71250.0	72500.0	73750.0	75000.0
50-60	3.0	85000.000000	5000.000000	80000.0	82500.0	85000.0	87500.0	90000.0

```
In [21]: data1 = {
    'age': [25, 30, 25, 40, 55, 60, 33, 28, 45, 50, 25, 30, 28, 30, 25],
    'income': [50000, 60000, 50000, 70000, 80000, 90000, 65000, 62000, 75000, 85000, 50000, 60000, 50000, 70000, 80000],
    'age_group': ['20-30', '30-40', '20-30', '40-50', '50-60', '50-60', '30-40', '20-30', '40-50', '50-60', '20-30', '30-40', '20-30', '40-50', '50-60']
}
```

```
# Define data in DataFrame
df = pd.DataFrame(data1)
```

```
In [22]: df.groupby('age_group')['income'].describe()
```

```
Out[22]:
```

	count	mean	std	min	25%	50%	75%	max
age_group								
20-30	6.0	58166.666667	10127.520262	50000.0	50000.0	56000.0	62000.0	75000.0
30-40	4.0	63750.000000	4787.135539	60000.0	60000.0	62500.0	66250.0	70000.0
40-50	2.0	72500.000000	3535.533906	70000.0	71250.0	72500.0	73750.0	75000.0
50-60	3.0	85000.000000	5000.000000	80000.0	82500.0	85000.0	87500.0	90000.0

```
In [23]: df.groupby('age_group')['income'].median()
```

```
Out[23]: age_group
20-30    56000.0
30-40    62500.0
40-50    72500.0
50-60    85000.0
Name: income, dtype: float64
```

```
In [24]: print("Column Names:", df.columns)
```

```
Column Names: Index(['age', 'income', 'age_group'], dtype='object')
```

```
In [27]: m_age = df['age'].mode()
print(m_age.values)
m_income = df['income'].mode()
print(m_income)
```

```
[25]
0    50000
Name: income, dtype: int64
```

```
In [30]: grouped = df.groupby('age_group')
income_list = grouped['income'].apply(list)
print(income_list)
```

```
age_group
20-30    [50000, 50000, 62000, 50000, 62000, 75000]
30-40              [60000, 65000, 60000, 70000]
40-50                  [70000, 75000]
50-60              [80000, 90000, 85000]
Name: income, dtype: object
```

```
In [31]: # PS2
```

```
In [2]: df = pd.read_csv('iris.csv')
```

```
In [3]: df
```

Out[3]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

150 rows × 5 columns

```
In [6]: grouped = df.groupby('variety')
for name, group in grouped:
    print(f"Species : {name}")
    print(group.describe())
# print(grouped.describe())

# print(df.groupby('variety').describe())
```

Species : Setosa

	sepal.length	sepal.width	petal.length	petal.width
count	50.00000	50.000000	50.000000	50.000000
mean	5.00600	3.428000	1.462000	0.246000
std	0.35249	0.379064	0.173664	0.105386
min	4.30000	2.300000	1.000000	0.100000
25%	4.80000	3.200000	1.400000	0.200000
50%	5.00000	3.400000	1.500000	0.200000
75%	5.20000	3.675000	1.575000	0.300000
max	5.80000	4.400000	1.900000	0.600000

Species : Versicolor

	sepal.length	sepal.width	petal.length	petal.width
count	50.000000	50.000000	50.000000	50.000000
mean	5.936000	2.770000	4.260000	1.326000
std	0.516171	0.313798	0.469911	0.197753
min	4.900000	2.000000	3.000000	1.000000
25%	5.600000	2.525000	4.000000	1.200000
50%	5.900000	2.800000	4.350000	1.300000
75%	6.300000	3.000000	4.600000	1.500000
max	7.000000	3.400000	5.100000	1.800000

Species : Virginica

	sepal.length	sepal.width	petal.length	petal.width
count	50.00000	50.000000	50.000000	50.00000
mean	6.58800	2.974000	5.552000	2.02600
std	0.63588	0.322497	0.551895	0.27465
min	4.90000	2.200000	4.500000	1.40000
25%	6.22500	2.800000	5.100000	1.80000
50%	6.50000	3.000000	5.550000	2.00000
75%	6.90000	3.175000	5.875000	2.30000
max	7.90000	3.800000	6.900000	2.50000

```
In [45]: percentile = df.groupby('variety').quantile([0.25,0.5,0.75])
```

```
In [56]: print(grouped.describe())
print(percentile)
```

	sepal.length								
	count	mean	std	min	25%	50%	75%	max	\
variety									
Setosa	50.0	5.006	0.352490	4.3	4.800	5.0	5.2	5.8	
Versicolor	50.0	5.936	0.516171	4.9	5.600	5.9	6.3	7.0	
Virginica	50.0	6.588	0.635880	4.9	6.225	6.5	6.9	7.9	

	sepal.width		...	petal.length		petal.width			\
	count	mean	...	75%	max	count	mean		
variety			...						
Setosa	50.0	3.428	...	1.575	1.9	50.0	0.246		
Versicolor	50.0	2.770	...	4.600	5.1	50.0	1.326		
Virginica	50.0	2.974	...	5.875	6.9	50.0	2.026		

	std	min	25%	50%	75%	max
variety						
Setosa	0.105386	0.1	0.2	0.2	0.3	0.6
Versicolor	0.197753	1.0	1.2	1.3	1.5	1.8
Virginica	0.274650	1.4	1.8	2.0	2.3	2.5

[3 rows x 32 columns]

		sepal.length	sepal.width	petal.length	petal.width
variety					
Setosa	0.25	4.800	3.200	1.400	0.2
	0.50	5.000	3.400	1.500	0.2
	0.75	5.200	3.675	1.575	0.3
Versicolor	0.25	5.600	2.525	4.000	1.2
	0.50	5.900	2.800	4.350	1.3
	0.75	6.300	3.000	4.600	1.5
Virginica	0.25	6.225	2.800	5.100	1.8
	0.50	6.500	3.000	5.550	2.0
	0.75	6.900	3.175	5.875	2.3

```
In [57]: # Group the data by variety; Select sepal.width column for each of the groups creat
summary_stats_species = df.groupby('variety')['sepal.width'].describe()

print("\nSummary Statistics by Species for Sepal Width:")
print(summary_stats_species)
```

Summary Statistics by Species for Sepal Width:

	count	mean	std	min	25%	50%	75%	max
variety								
Setosa	50.0	3.428	0.379064	2.3	3.200	3.4	3.675	4.4
Versicolor	50.0	2.770	0.313798	2.0	2.525	2.8	3.000	3.4
Virginica	50.0	2.974	0.322497	2.2	2.800	3.0	3.175	3.8

```
In [7]: # Group by variety and compute the median for numeric columns
# median_values =
df.groupby('variety').median()

# print("Median Values by Species:")
# print(median_values)
```

Out[7]:

	sepal.length	sepal.width	petal.length	petal.width
variety				
Setosa	5.0	3.4	1.50	0.2
Versicolor	5.9	2.8	4.35	1.3
Virginica	6.5	3.0	5.55	2.0

```
In [59]: # Group the data by variety; Select sepal.width column for each of the groups creat
median_sepal_length = df.groupby('variety')['sepal.length'].median()
print("Median Sepal Length by Species:")
print(median_sepal_length)
```

Median Sepal Length by Species:

variety

Setosa 5.0

Versicolor 5.9

Virginica 6.5

Name: sepal.length, dtype: float64

```
In [60]: # Calculate & print mode for sepal.width
mode_width = df['sepal.width'].mode()
print(f"Mode of Width: {mode_width.values}")
```

Mode of Width: [3.]

In []: